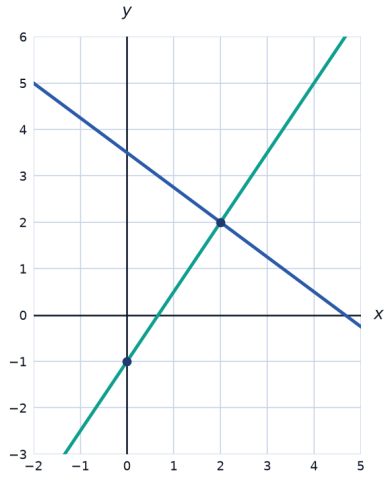
# Continuity and the Intermediate Value Theorem

## Section 1: Three-Part Definition of Continuity

**1** Which three conditions must be satisfied for a function $f$ to be continuous at $x = c$?




**a.** $f(c)$ is defined, $\lim_{x \to c} f(x)$ exists, and $\lim_{x \to c} f(x) = f(c)$

**b.** $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

**c.** $f(c)$ is continuous on $(a, b)$, $f(c) = 0$, and $f'(c) \neq 0$

**d.** $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

**2** Let $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$. What value must be assigned to $f(3)$ to make $f$ continuous at $x = 3$?

**a.** $3$

**b.** $6$

**c.** $0$

**d.** No value can make it continuous

**3** The function $g(x) = \frac{1}{x - 2}$ has what type of discontinuity at $x = 2$?

**a.** Infinite (essential) discontinuity

**b.** Removable discontinuity

**c.** Jump discontinuity

**d.** Oscillating discontinuity

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## Section 2: Piecewise Functions & Continuity

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**4** For what value of $k$ is $f(x) = kx^2 + 1$ for $x \leq 2$, and $f(x) = 3x - 1$ for $x > 2$ continuous at $x = 2$?

**a.** 1

**b.** 2

**c.** 3

**d.** 0

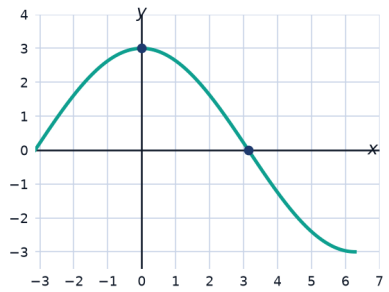
**5** If $g(x) = \frac{\sin(ax)}{x}$ for $x \neq 0$, and $g(0) = 4$, what value of $a$ ensures continuity at $x = 0$?

**a.** 1

**b.** 2

**c.** 4

**d.** $\frac{1}{4}$




**6** The function $h(x) = x + 2$ for $x < 1$, and $h(x) = 2x + 1$ for $x \geq 1$ has what property at $x = 1$?

**a.** Continuous because both one-sided limits equal 3

**b.** Jump discontinuity of size 1

**c.** Infinite discontinuity

**d.** Undefined at $x = 1$

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## Section 3: Intermediate Value Theorem (IVT)

**7** What are the hypotheses of the Intermediate Value Theorem for a function $f$ on interval $[a, b]$?

**a.** $f$ must be continuous on the closed interval $[a, b]$

**b.** $f$ must be differentiable on $(a, b)$

**c.** $f(a)$ and $f(b)$ must have opposite signs

**d.** $f$ must be strictly increasing on $[a, b]$

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**8** Let $f$ be continuous on $[1, 5]$ with $f(1) = -3$ and $f(5) = 7$. Does the IVT guarantee a solution to $f(c) = 0$ for some $c \in (1, 5)$?

**a.** Yes, because $0$ lies between $-3$ and $7$

**b.** No, $f$ must also be differentiable

**c.** No, $f$ must be a polynomial

**d.** Only if $f'(x) > 0$ for all $x$

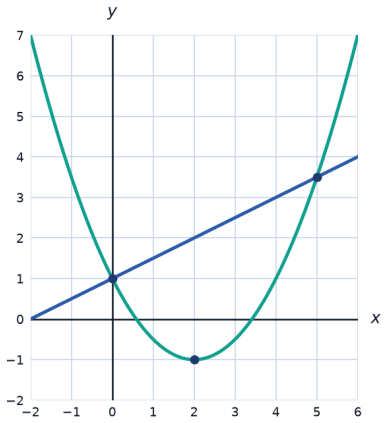
**9** On which interval is the polynomial $P(x) = x^3 - 3x - 1$ guaranteed to have a real zero by the IVT?

**a.** $[1, 2]$

**b.** $[0, 1]$

**c.** $[-1, 0]$

**d.** $[2, 3]$




## Section 4: Extreme Value Theorem (EVT)

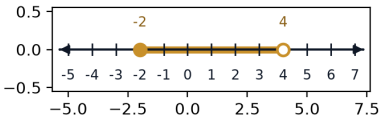
**10** What does the Extreme Value Theorem guarantee for a function $f$ continuous on a closed interval $[a, b]$?

**a.** $f$ attains both an absolute maximum and an absolute minimum on $[a, b]$

**b.** $f$ has a horizontal tangent on $(a, b)$

**c.** $f'(c) = 0$ for some $c \in (a, b)$

**d.** $f$ is invertible on $[a, b]$




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**11** Why does the function $f(x) = \frac{1}{x}$ NOT attain an absolute maximum on $(0, 2]$?

**a.** The interval is open at $0$ and $f$ increases without bound as $x \to 0^+$

**b.** $f$ is not differentiable on $(0, 2]$

**c.** $f$ is bounded on $(0, 2]$

**d.** $f(2) = 0.5$ is the maximum

**12** Find the absolute maximum value of $f(x) = -x^2 + 4x + 1$ on the closed interval $[0, 3]$.

**a.** $5$

**b.** $4$

**c.** $1$

**d.** $6$

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